

PROJECT VESPER — Simple Tech Stack Guide

What is used, Why it is used, Where it is used & Why alternatives were rejected

1. FRONTEND & MAP CONSOLE

Vanilla CSS3 (IRCTC Color Theme)

- **What is it?** Plain CSS styling using official government colors (#213d77 Navy, #fb792b Orange).
- **Why is it used?** Loads instantly, high contrast for 24/7 control rooms, 100% compliant with government rules.
- **Where is it used in VESPER?** Command Center control dashboard styling.
- **Substitute:** TailwindCSS / Bootstrap.
- **Why substitute was NOT used:** Heavy CSS frameworks add bloat and make dashboards look like generic commercial websites.

Vanilla JavaScript & SVG Icons

- **What is it?** Pure JavaScript with clean vector SVG icons (zero emojis/unicode characters).
- **Why is it used?** Zero lag during fast live updates and 100% consistent icon rendering on any police screen.
- **Where is it used in VESPER?** Dashboard buttons, tabs, live updates, and panel docking logic.
- **Substitute:** React.js / FontAwesome / Emojis.
- **Why substitute was NOT used:** React adds rendering delay during fast map updates. Emojis show up broken on defense/police hardware.

Leaflet.js + OpenStreetMap

- **What is it?** Free open-source map engine and road vector data.
- **Why is it used?** Shows smooth 60 FPS live maps with vehicle paths at ZERO licensing cost.
- **Where is it used in VESPER?** Central Tactical GIS map screen.
- **Substitute:** Google Maps API / Mapbox.
- **Why substitute was NOT used:** Google Maps charges per map load, costing cities Rs 4+ Lakhs/month. Leaflet is 100% free.

2. AI VISION & EDGE VIDEO PROCESSING

YOLOv8-Nano

- **What is it?** Super-fast AI model for detecting vehicle number plates and car types.
- **Why is it used?** Runs extremely fast (45+ frames per second) even on low-cost hardware.
- **Where is it used in VESPER?** Edge AI camera nodes on public buses and street poles.
- **Substitute:** Faster R-CNN / SSD.
- **Why substitute was NOT used:** Faster R-CNN is too slow (~10 FPS). SSD misses small or high-angle number plates.

PaddleOCR v4 + STN Dewarping

- **What is it?** AI text reader that automatically straightens crooked license plate photos.
- **Why is it used?** Reads English + Hindi plates with >93% accuracy, even from moving buses on bumpy roads.
- **Where is it used in VESPER?** Edge camera software right after license plate detection.
- **Substitute:** Tesseract OCR / EasyOCR.
- **Why substitute was NOT used:** Tesseract is built for flat scanned paper; fails completely (<50%) on moving road photos.

NVIDIA Jetson + TensorRT (Edge AI)

- **What is it?** Compact, low-power AI mini-computer mounted inside buses and camera poles.
- **Why is it used?** Processes video right at the camera, saving Rs 50+ Lakhs/month in 4G data cost.
- **Where is it used in VESPER?** Inside public buses (DTC/BMTC) and CCTV poles.
- **Substitute:** Raspberry Pi / Cloud Video Streaming.
- **Why substitute was NOT used:** Raspberry Pi lacks AI hardware (drops to 3 FPS). Cloud video streaming wastes huge 4G bandwidth.

3. TRACKING & TRAJECTORY AI

ByteTrack + FastReID Engine

- **What is it?** AI that follows vehicles across video frames and matches car colors/shapes.
- **Why is it used?** Keeps tracking wanted cars even if the number plate is dirty, hidden, or unreadable.
- **Where is it used in VESPER?** Bus cameras (local tracking) and Cloud engine (tracking across camera blind spots).
- **Substitute:** DeepSORT.
- **Why substitute was NOT used:** DeepSORT overworks the computer on every single frame, causing lag on camera hardware.

Hidden Markov Model (HMM) Map Matching

- **What is it?** Smart math algorithm that connects camera sightings along real road maps.
- **Why is it used?** Predicts a fleeing vehicle's exact route so police can set up roadblocks ahead.
- **Where is it used in VESPER?** Cloud server whenever a wanted vehicle is spotted.
- **Substitute:** Straight-line drawing / Deep Learning AI (LSTMs).
- **Why substitute was NOT used:** Straight lines ignore roads (predicting driving through lakes/buildings). Deep AI gives unpredictable guesswork.

4. BACKEND & DATA INFRASTRUCTURE

FastAPI & Node.js Express

- **What is it?** Fast asynchronous web server software.
- **Why is it used?** Handles thousands of camera pings per second and streams live map data smoothly.
- **Where is it used in VESPER?** Main cloud backend servers.
- **Substitute:** Django / Flask.
- **Why substitute was NOT used:** Django and Flask block server threads, freezing under heavy camera traffic.

Redis In-Memory Database

- **What is it?** Super-fast memory database that checks plates instantly.
- **Why is it used?** Checks detected plates against police wanted lists in under 10 milliseconds.
- **Where is it used in VESPER?** Cloud server hotlist cache & instant alert dispatcher.
- **Substitute:** Standard SQL Database (PostgreSQL).
- **Why substitute was NOT used:** Standard SQL databases read from hard drives, taking 50-100ms and creating alert delays.

SQLite (Store & Forward)

- **What is it?** Lightweight offline database stored directly inside the edge computer.
- **Why is it used?** Saves sightings when buses pass through tunnels/dead zones and auto-syncs when back online.
- **Where is it used in VESPER?** Inside each bus and CCTV pole edge hardware.
- **Substitute:** Temporary JSON text files.
- **Why substitute was NOT used:** Text files corrupt easily when a bus engine shuts off unexpectedly. SQLite prevents data loss.

5. INTEGRATION & LEGAL COMPLIANCE

MoRTH AIS-140 Bus Telemetry Standard

- **What is it?** Govt of India mandatory standard for public bus GPS and emergency hardware.
- **Why is it used?** Lets VESPER plug directly into existing public buses with ZERO extra hardware cost.
- **Where is it used in VESPER?** Public bus fleet integration (DTC, BMTC, BEST).
- **Substitute:** Custom hardware wiring.
- **Why substitute was NOT used:** Installing custom hardware on 10,000 buses costs tens of crores. AIS-140 is already installed.

BSA Sec 63 SHA-256 & DPDP Act Face-Blurring

- **What is it?** Digital signature for legal evidence + automatic face blurring for citizen privacy.
- **Why is it used?** Makes photos 100% admissible in court while protecting pedestrian privacy automatically.
- **Where is it used in VESPER?** On edge camera nodes before photos are saved or sent.
- **Substitute:** Plain unhashed JPEG photos.
- **Why substitute was NOT used:** Plain photos can be rejected in court as tampered; manual face blurring takes too much human time.